# Energy Transport in Disordered Networks —

## From Transport Efficiency of Photosynthetic Light-Harvesting to Coherent Interactions in Frozen Rydberg Gases

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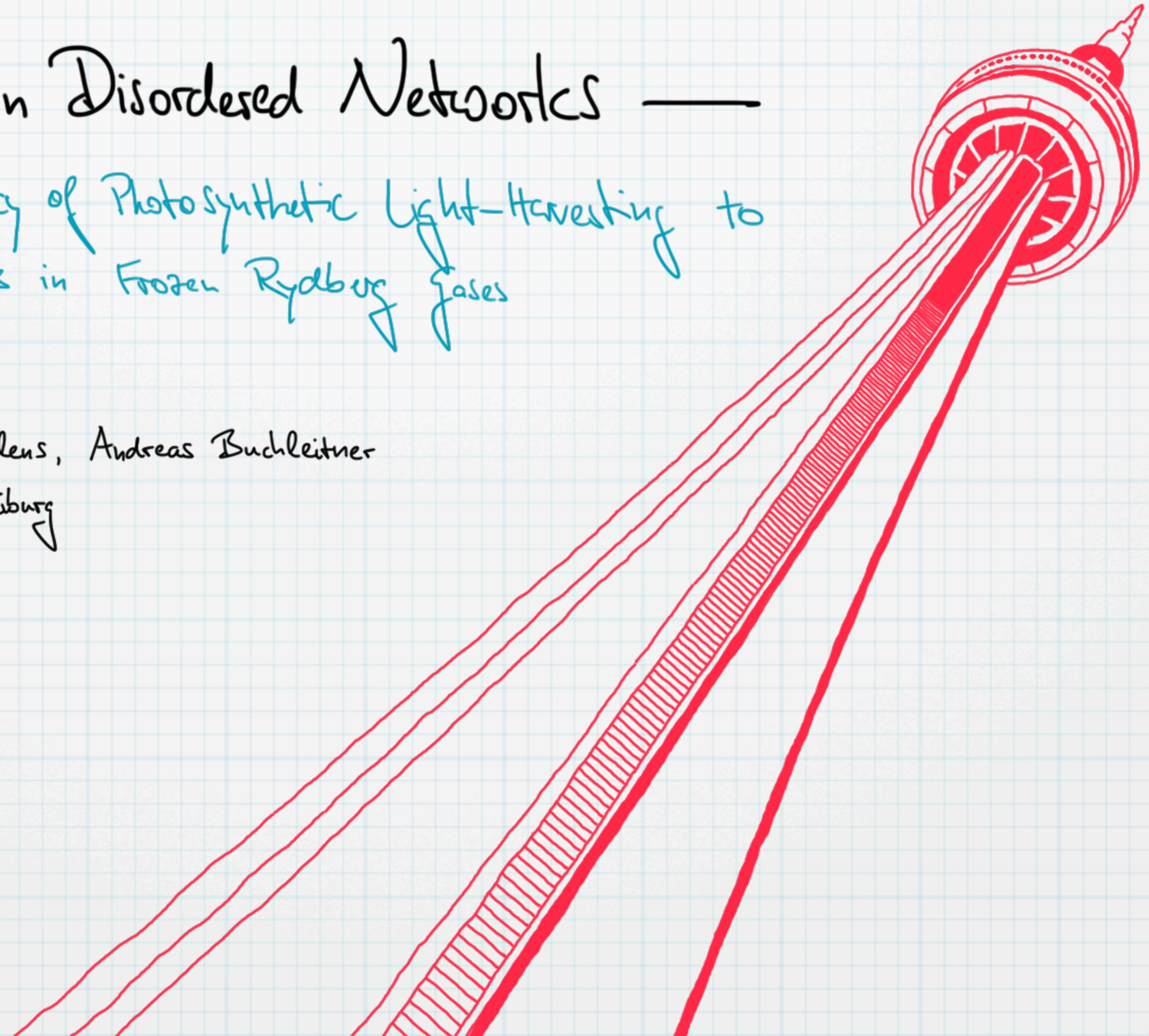
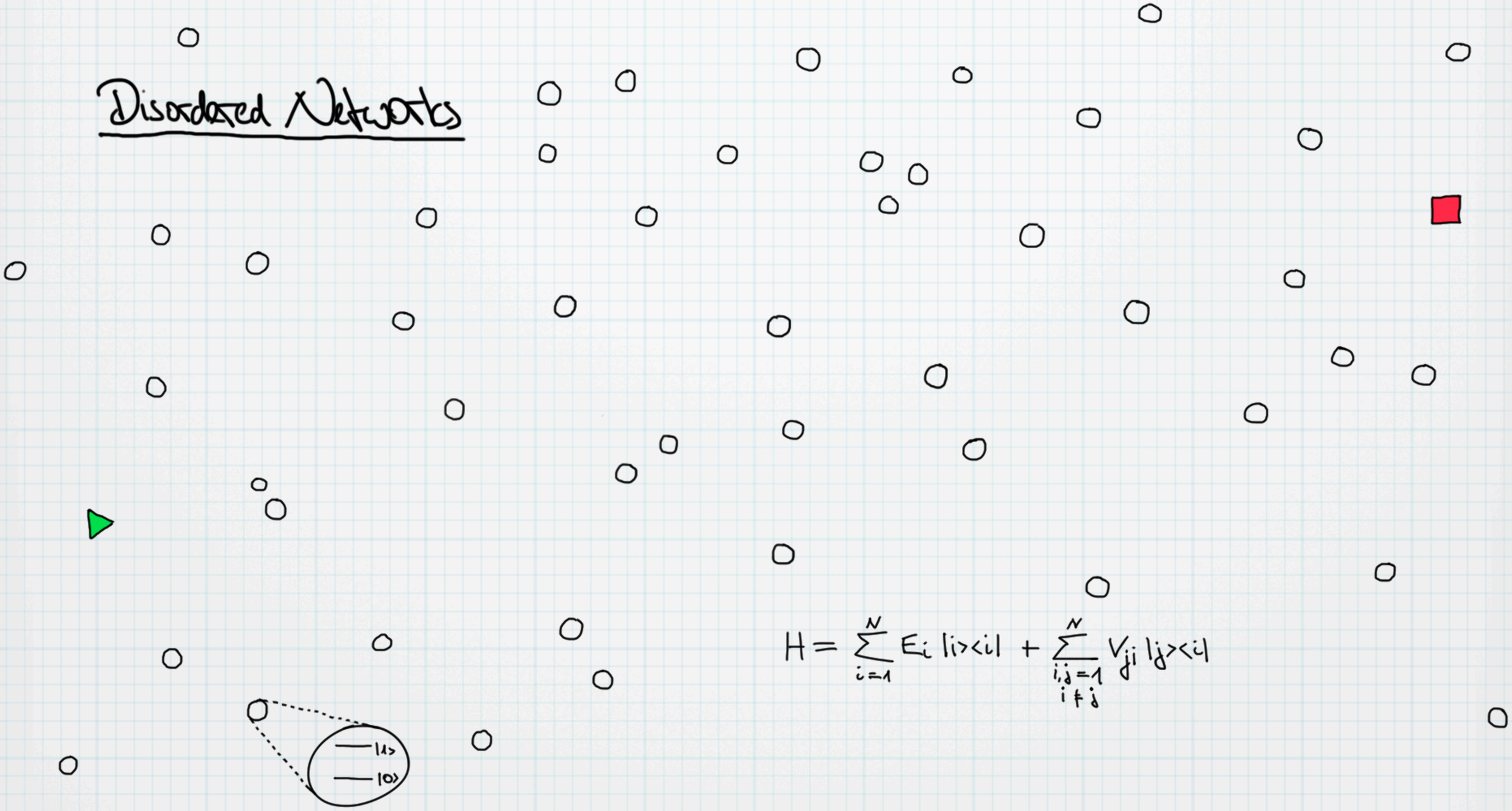


# Disordered Networks




$$H = \sum_{i=1}^{N} E_i \, |i\rangle\langle i| + \sum_{\substack{i,j=1 \\ i \neq j}}^{N} V_{ji} \, |j\rangle\langle i|$$

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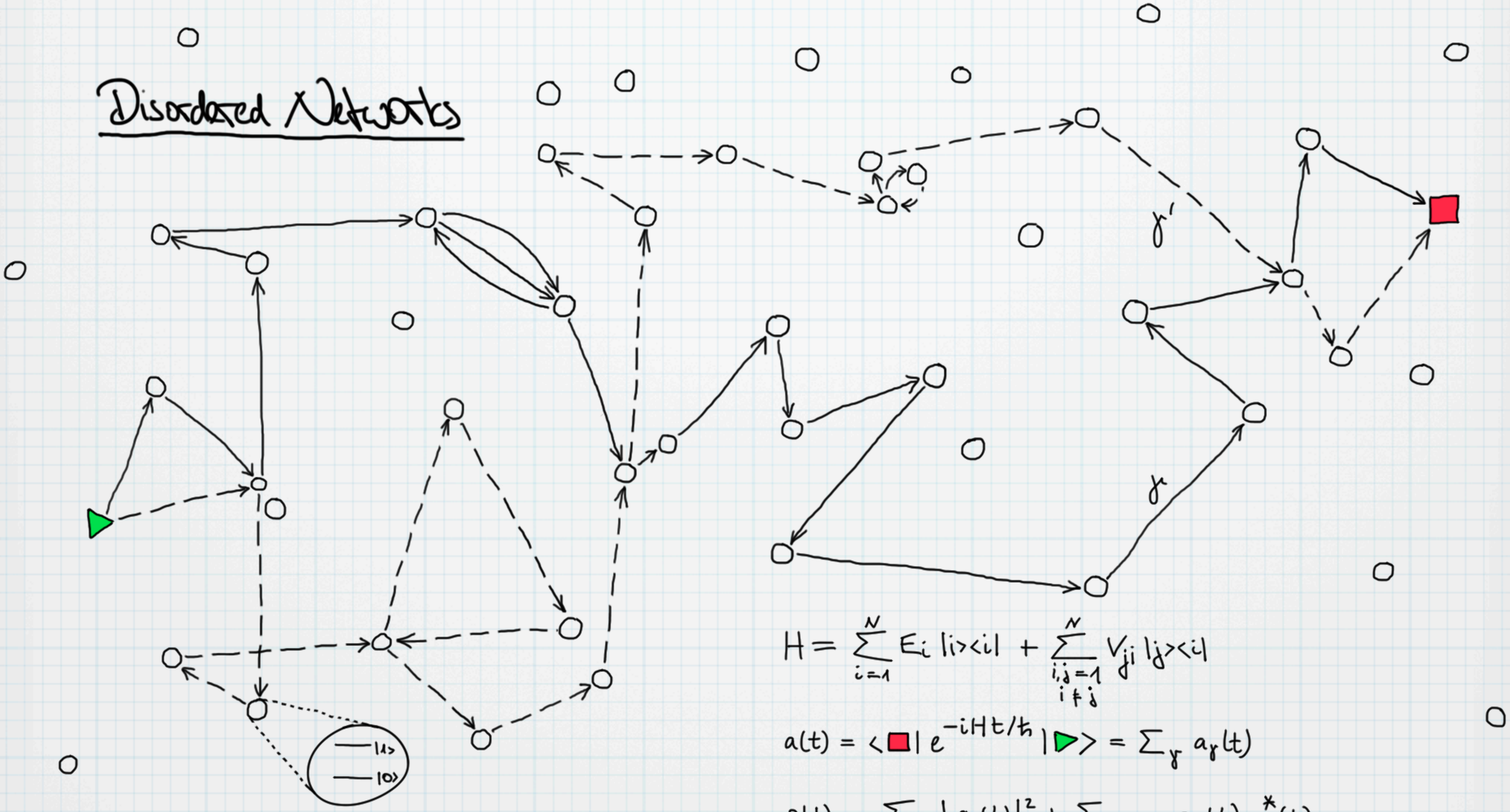



$$H = \sum_{i=1}^{N} E_i \, |i\rangle\langle i| + \sum_{\substack{i,j=1 \\ i \neq j}}^{N} V_{ji} \, |j\rangle\langle i|$$

$$a(t) = \langle \blacksquare | e^{-iHt/\hbar} | \blacktriangleright \rangle = \sum_{\gamma} a_{\gamma}(t)$$

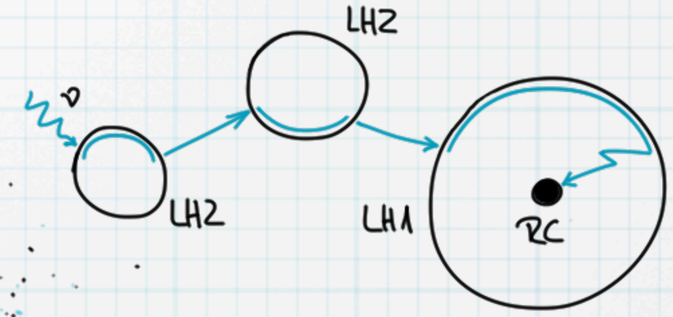
$$p(t) = \sum_{\gamma} |a_{\gamma}(t)|^2 + \sum_{\gamma \neq \gamma'} a_{\gamma}(t) \, a^*_{\gamma'}(t)$$

# Implementation?

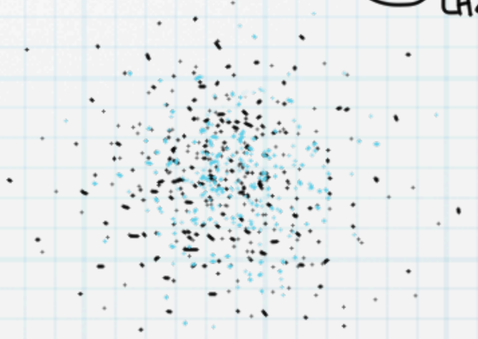
LARGE — small

$N = 7$ — Fenna–Matthews–Olson complex

$N = 10 \ldots 100$ — Light-Harvesting system 1, LH 2

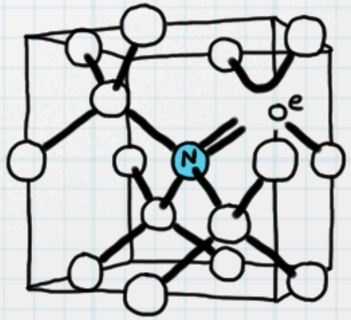



$N = 1 \ldots 10^5$ — frozen Rydberg gases



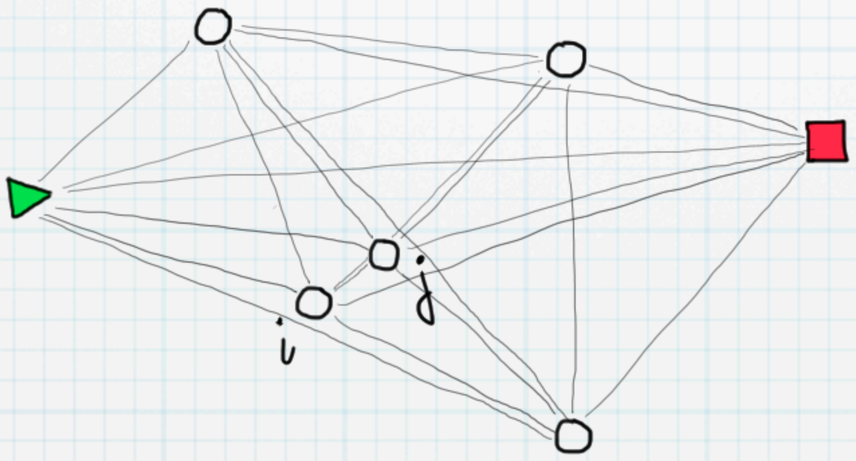
$\frac{N}{cm^3} > 10^{12}$ — impurity exciton band conduction in semiconductor alloys

Nitrogen Vacancy centres in diamond



# Small Networks

Energy Transport in Photosynthetic Light-Harvesting



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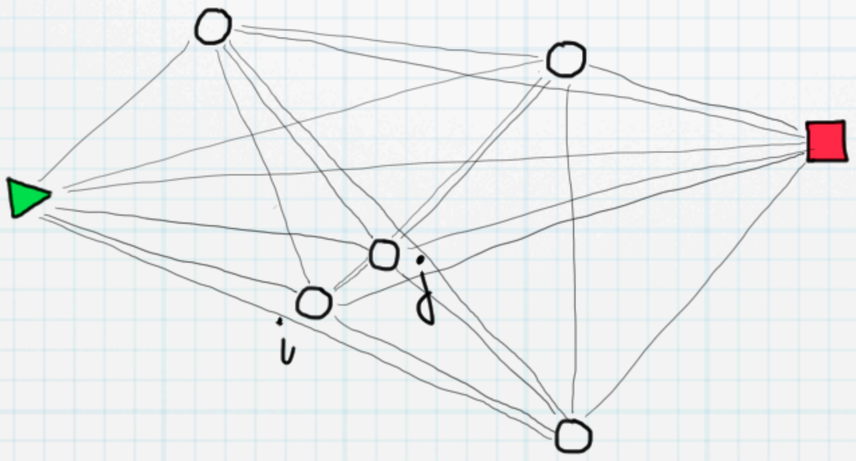



## Suppression of Transport

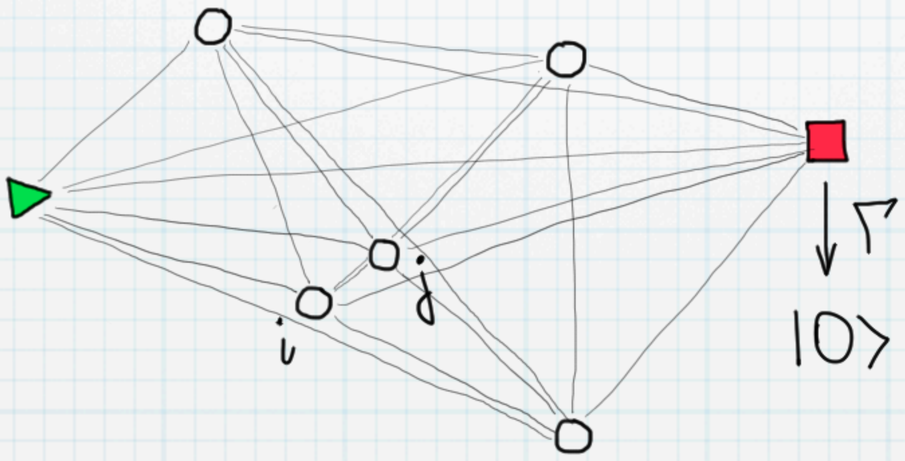
- pair formation

$$\dot{\rho} = \frac{1}{i\hbar}[H, \rho]$$

$$V_{ij} = \underbrace{\mathcal{A}(\vec{d}_i, \vec{d}_j, \vartheta_{ij}, \varphi_{ij})}_{\text{angular}} \underbrace{r_{ij}^{-3}}_{\text{radial}}$$

# Small Networks

Energy Transport in Photosynthetic Light-Harvesting




$$\dot{\rho} = \frac{1}{i\hbar}[H, \rho] + \mathcal{L}_{sink}(\rho)$$

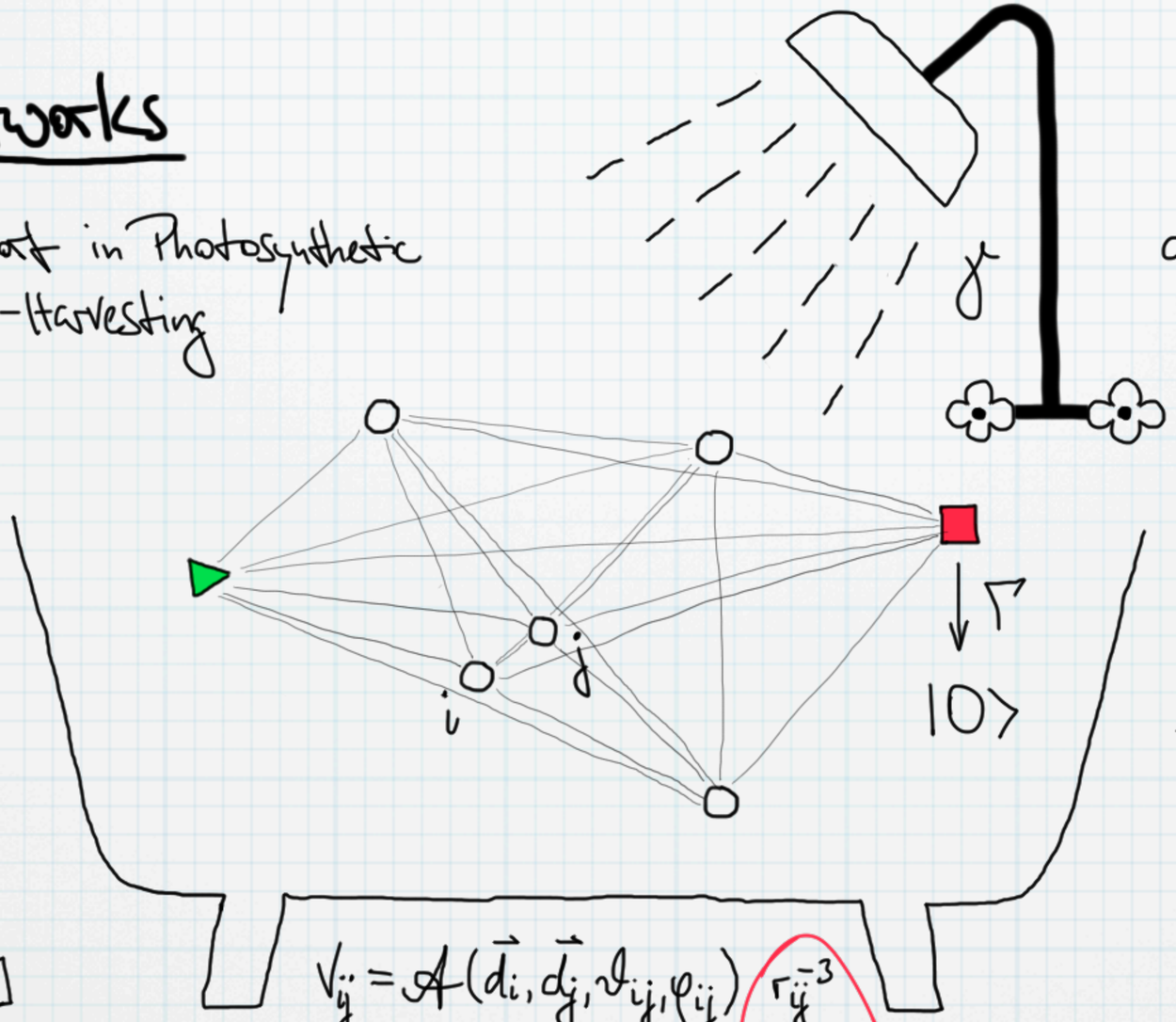
$$V_{ij} = \underbrace{\mathcal{A}(\vec{d}_i, \vec{d}_j, \vartheta_{ij}, \varphi_{ij})}_{\text{angular}} \underbrace{r_{ij}^{-3}}_{\text{radial}}$$

## Suppression of Transport

- pair formation
- destructive interference

# Small Networks

Energy Transport in Photosynthetic Light-Harvesting




dephasing, dissipation, ...

$$\dot{\rho} = \frac{1}{i\hbar}[H,\rho] + \mathcal{L}_{sink}(\rho) + \mathcal{L}_{bath}(\rho)$$

$$V_{ij} = \underbrace{\mathcal{A}(\vec{d}_i, \vec{d}_j, \vartheta_{ij}, \varphi_{ij})}_{\text{angular}} \underbrace{r_{ij}^{-3}}_{\text{radial}}$$

## Suppression of Transport

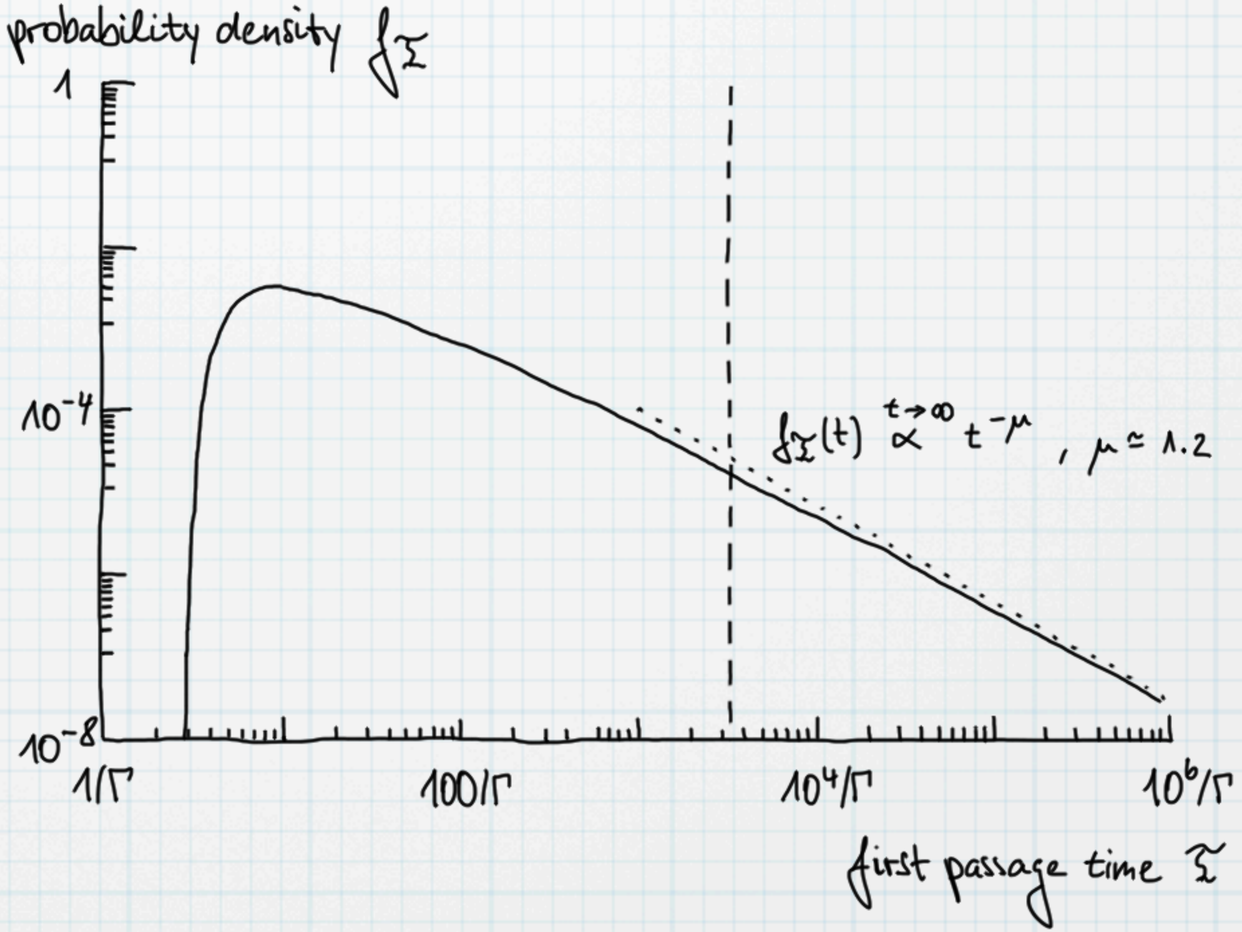
- pair formation
- destructive interference

# Transport Statistics

- timescales

  reference $T = \frac{\pi}{2\hbar |V_{\triangleright\square}|}$

  sink $\Gamma^{-1} = 0.1\,T$



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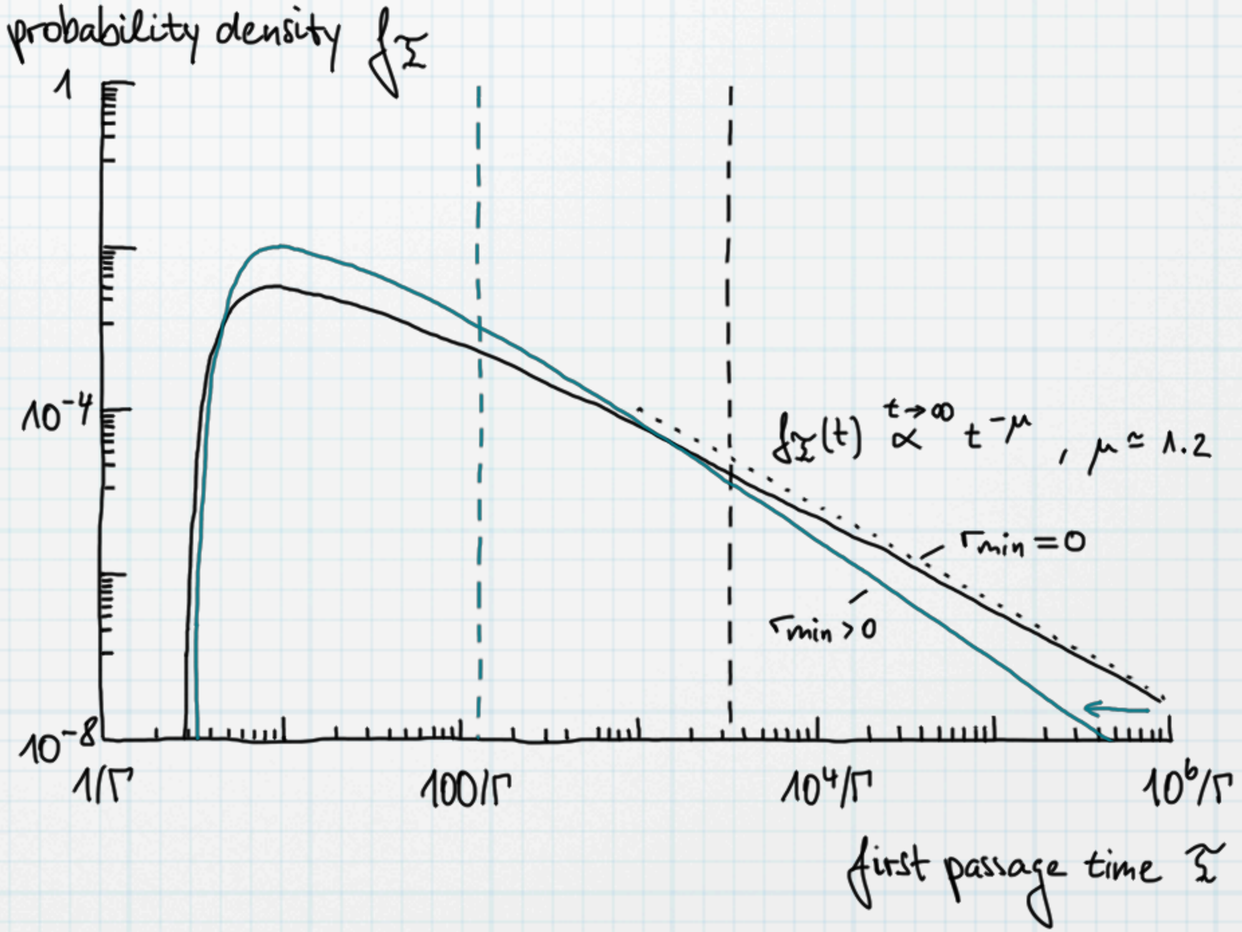
- $\Gamma_{min} > 0 \Rightarrow$ less pair localization



# Transport Statistics

- timescales

  reference $T = \frac{\pi}{2\hbar |V_{\triangleright\square}|}$

  sink $\Gamma^{-1} = 0.1\,T$

  dephasing $\gamma^{-1}$

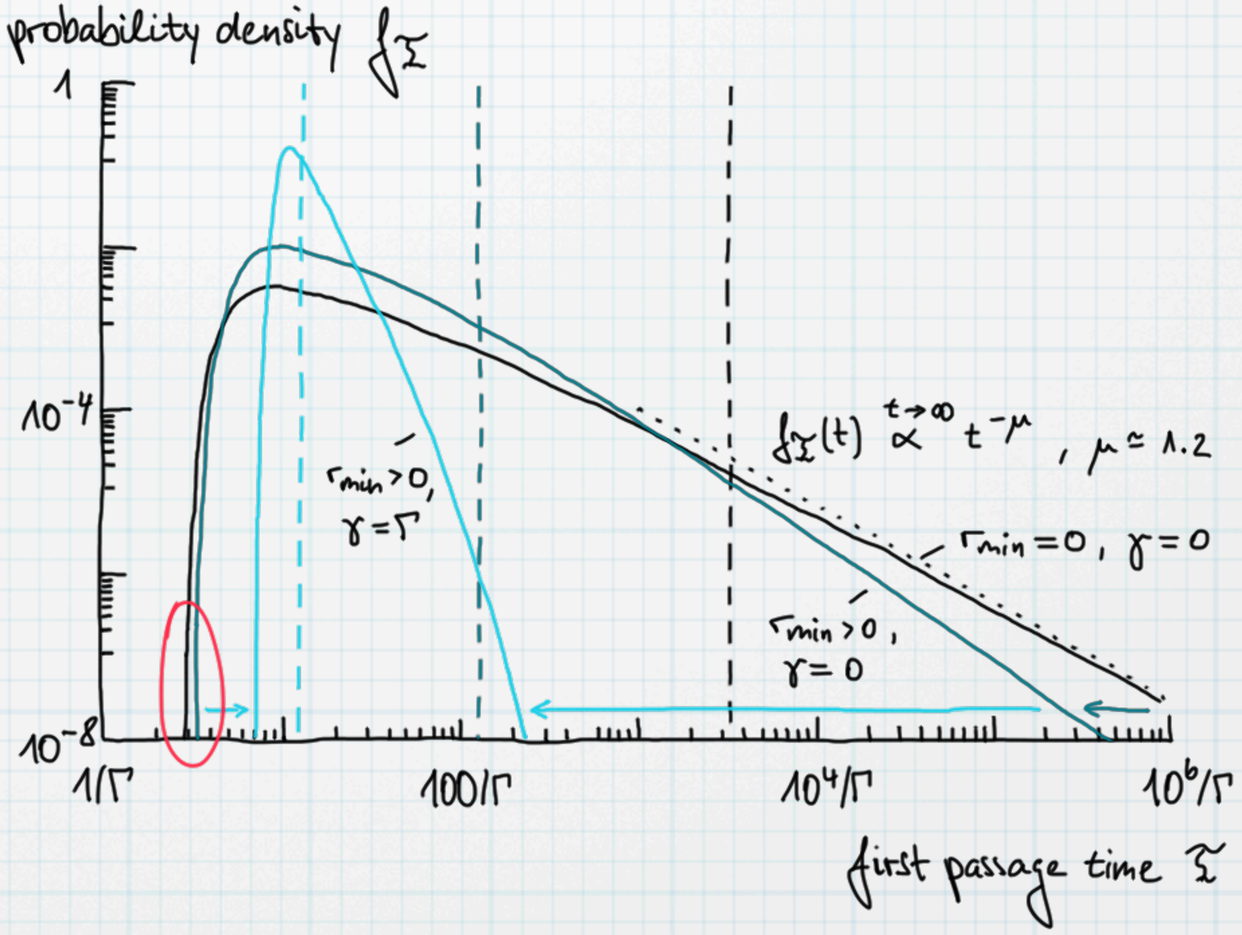
- $\Gamma_{min} > 0 \Rightarrow$ less pair localization

- $\gamma > 0 \Rightarrow$ less quantum interference,

  coherent $\xrightarrow{\text{rate } \gamma}$ incoherent transport

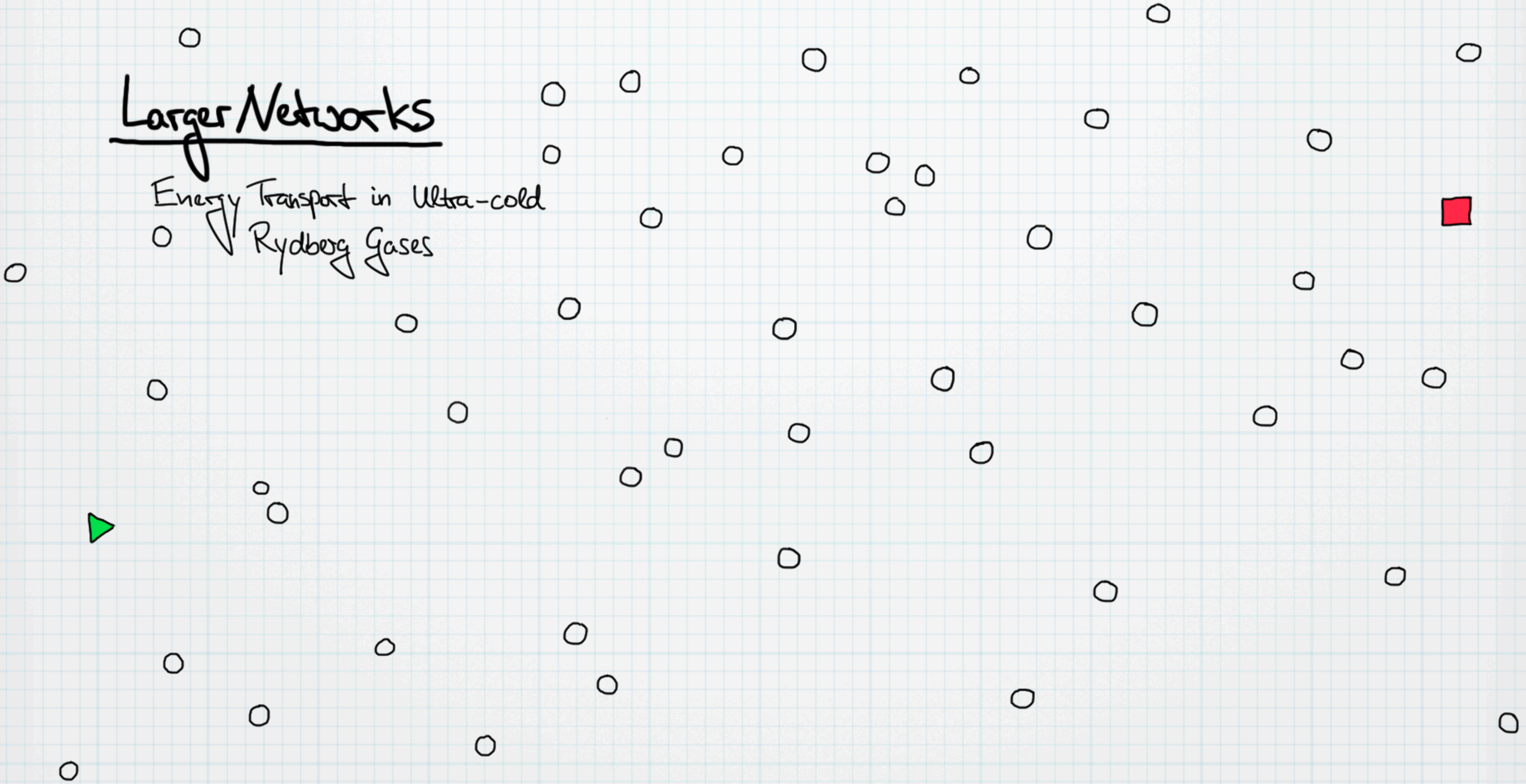
- optimized coherent $>$ noise-assisted transport

  ($\rightsquigarrow$ T. Zech)



# Larger Networks

Energy Transport in Ultra-cold Rydberg Gases



# Rydberg Atoms

- atoms in highly excited states

  $n = 50 \ldots 500$

- radius $\sim n^2$

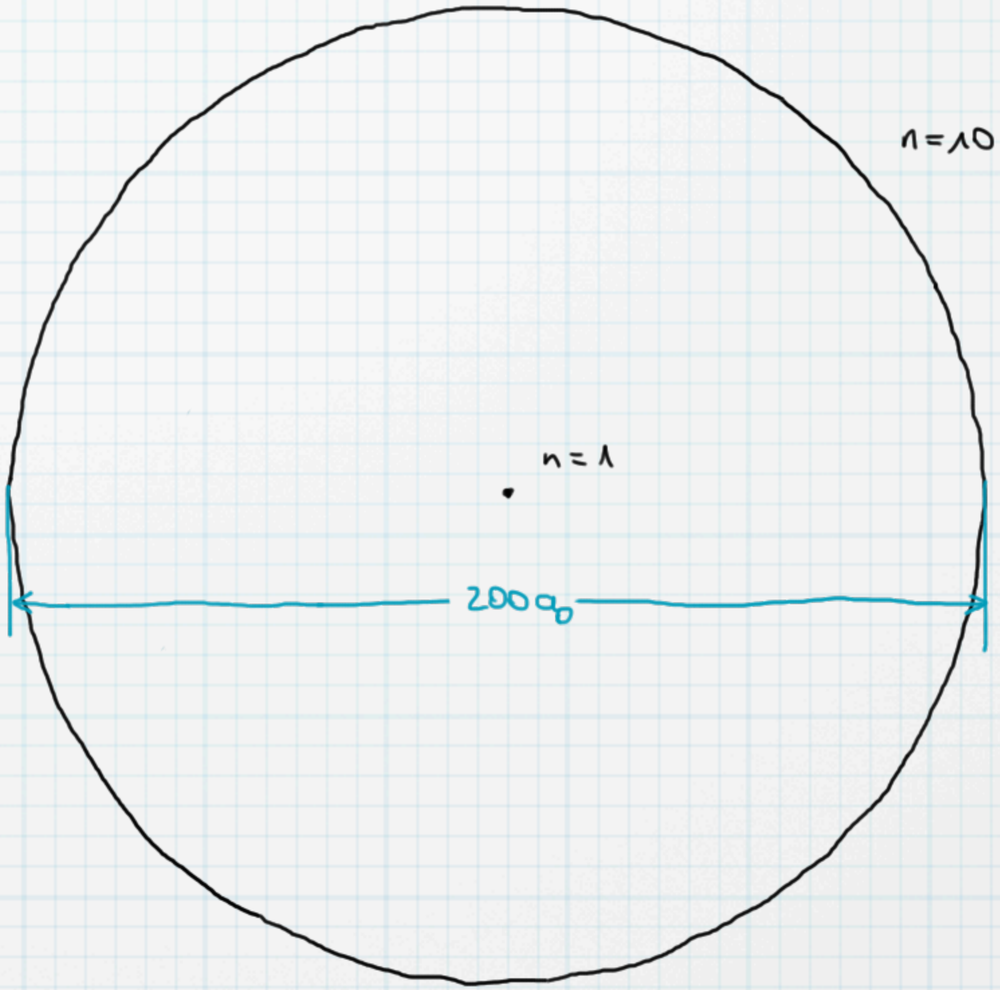
  e.g. 1 µm for hydrogen & $n = 137$

- very strong interactions

  polarizability $\sim n^7$

- long-lived $\sim n^3$

  however fragile



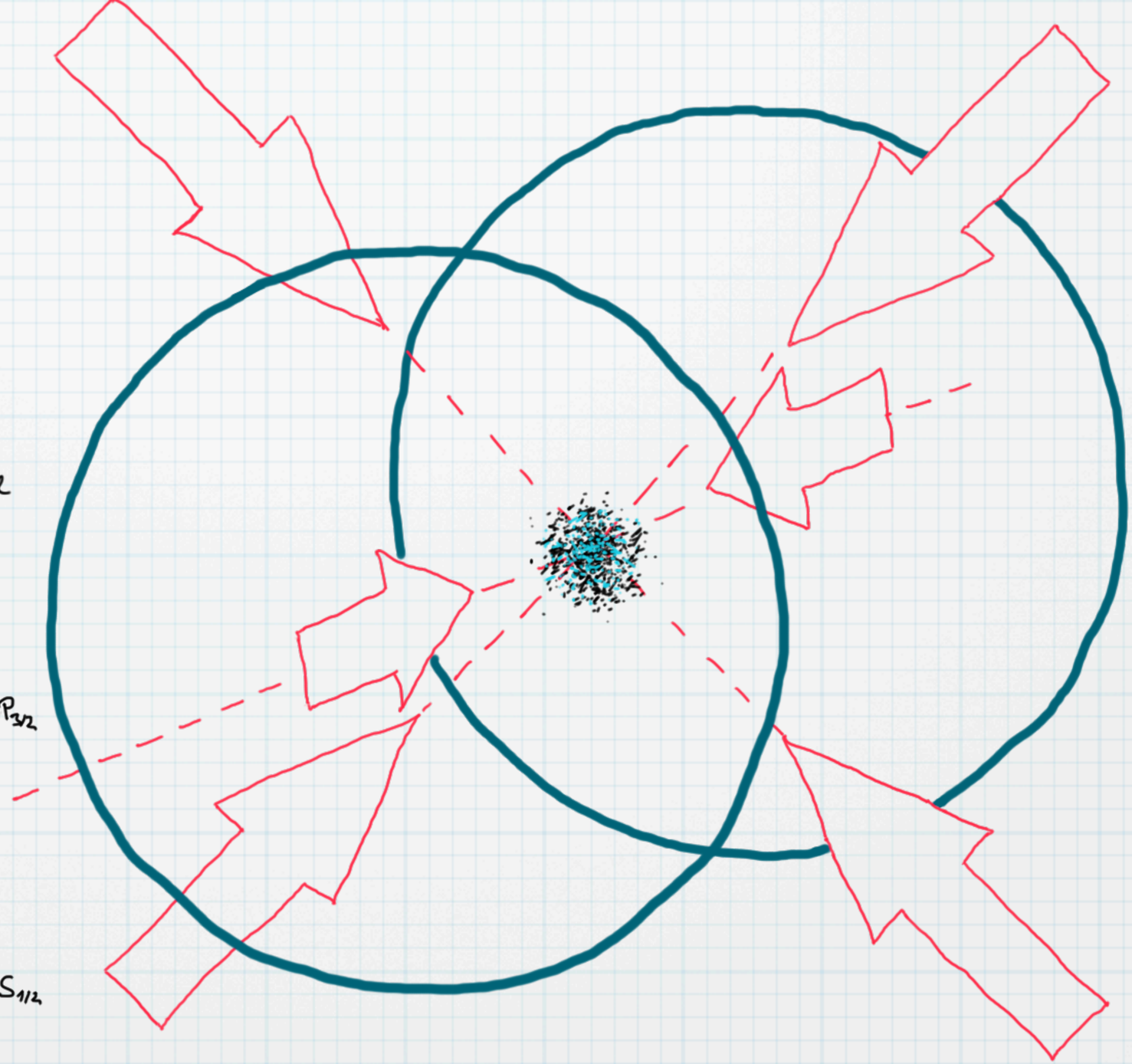
# Rydberg Gases I

① load a MOT with
+ Rubidium vapor

② cool down to
+ $T < 100\,\mu K$

③ excite to
Rydberg states

$N \sim 10^4$ frozen
Rydberg atoms
$N/V \sim 10^7/cm^3$

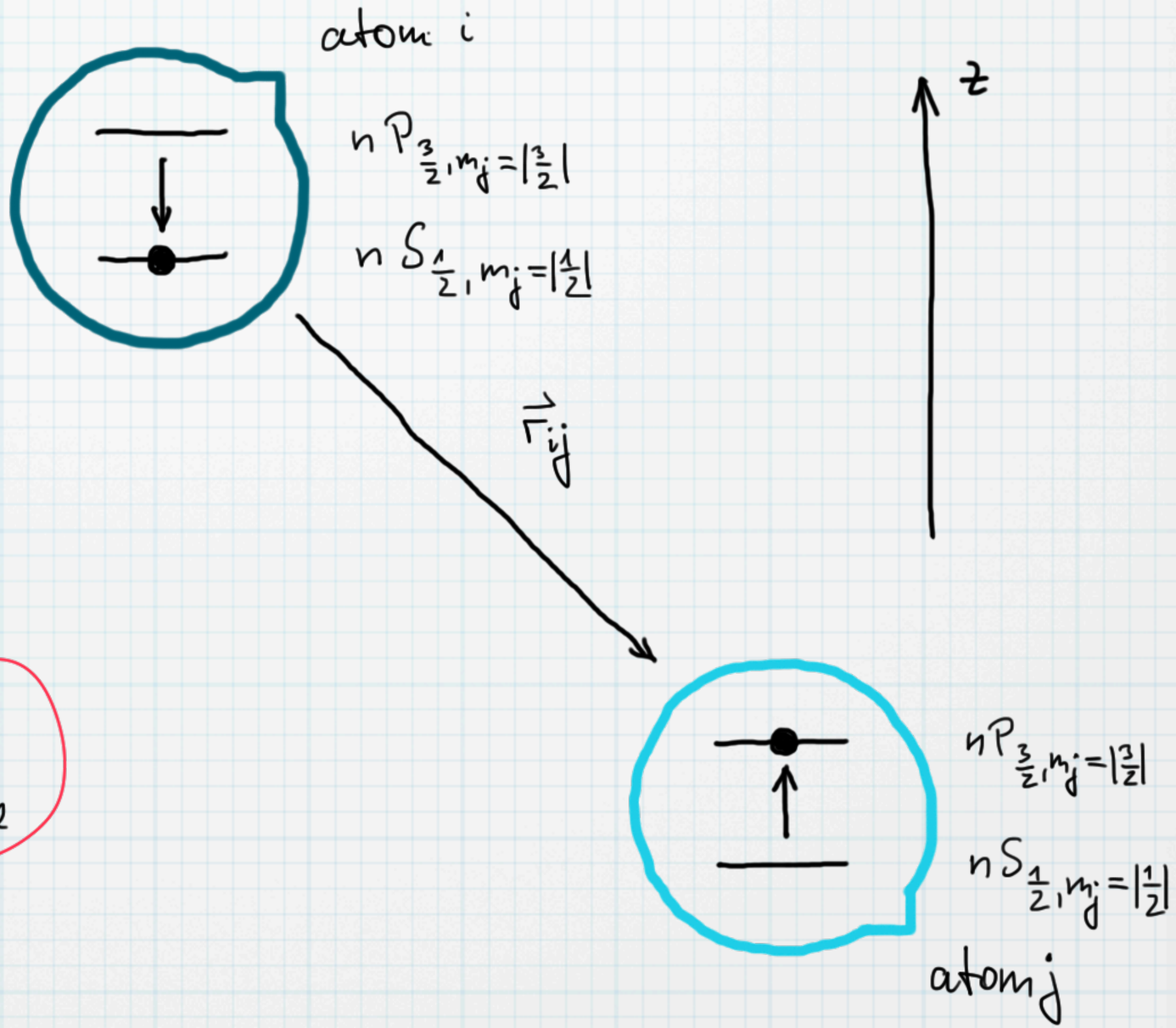


# Rydberg Gases II

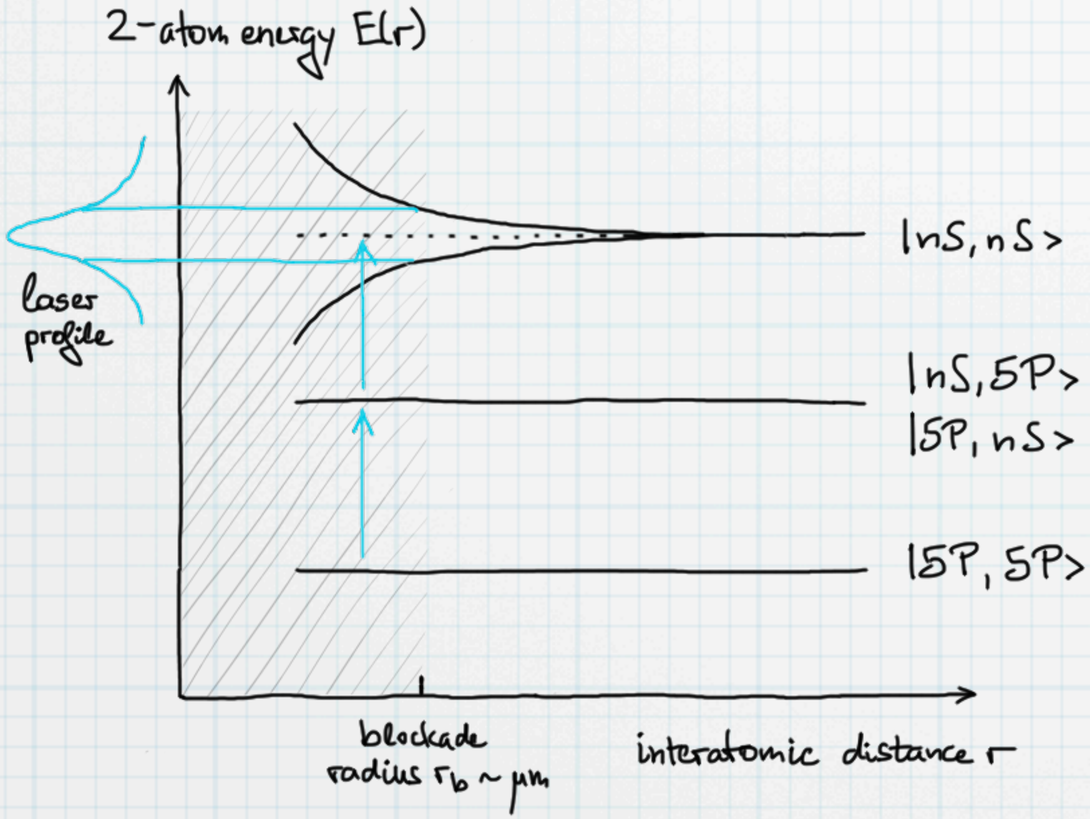
- resonant non-radiative energy transfer between the nP and nS states of different Rydberg atoms

- dipole-dipole interaction

$$V_{ij} \propto \underbrace{\left(\cos^2\vartheta_{ij} - \frac{1}{3}\right)}_{\text{angular}} \underbrace{r_{ij}^{-3}}_{\text{radial}}$$

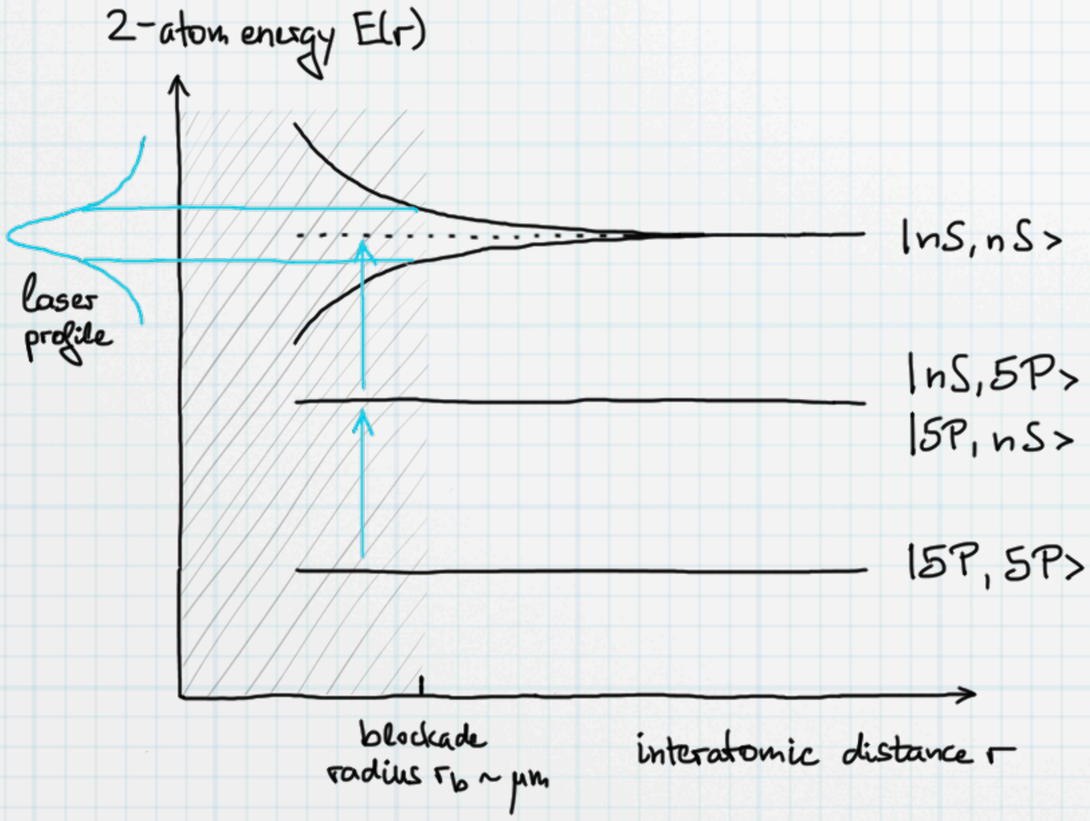


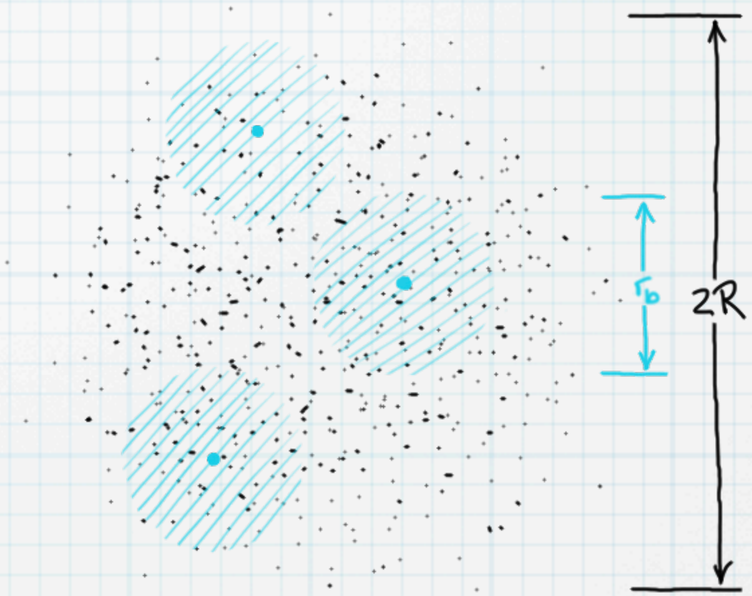
# Dipole Blockade

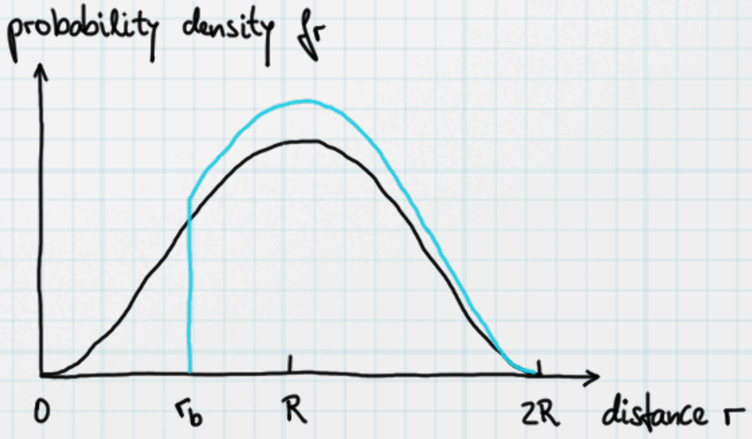


# Dipole Blockade

$$\frac{1}{\sqrt{W_b}} \sum_m e^{i\vec{k}\cdot\vec{r}_m} |5P, \ldots, \overset{m\text{-th atom}}{nS}, \ldots, 5P\rangle$$



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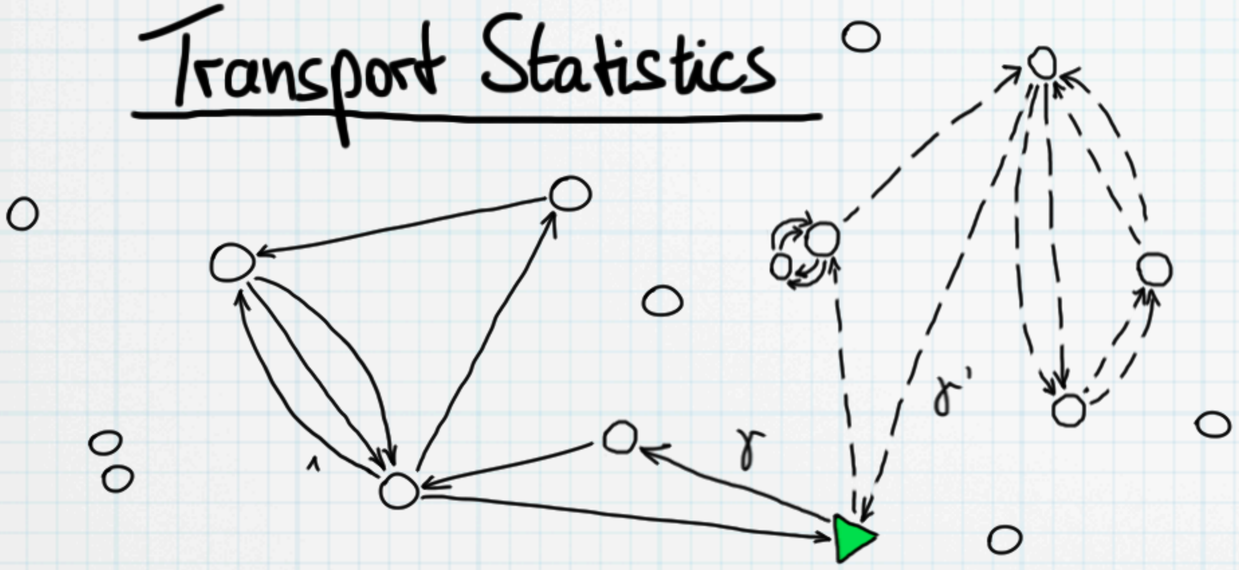
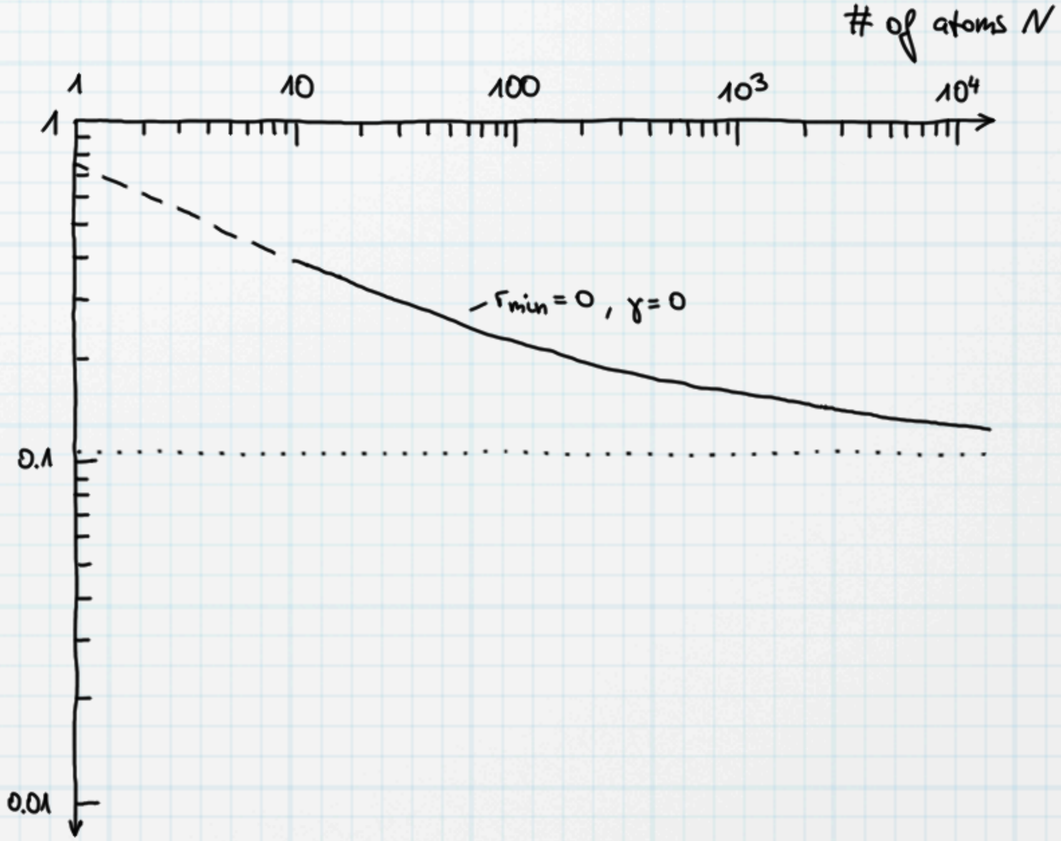



asymptotic ensemble-averaged return probability $\overline{P_0}$

- return probability ($\gamma = 0$)

$$\overline{P_0} = \overline{\lim_{t\to\infty} \frac{1}{t} \int_0^t |\langle \triangleright | e^{-iHt'/\hbar} | \triangleright \rangle|^2 \, dt'}$$

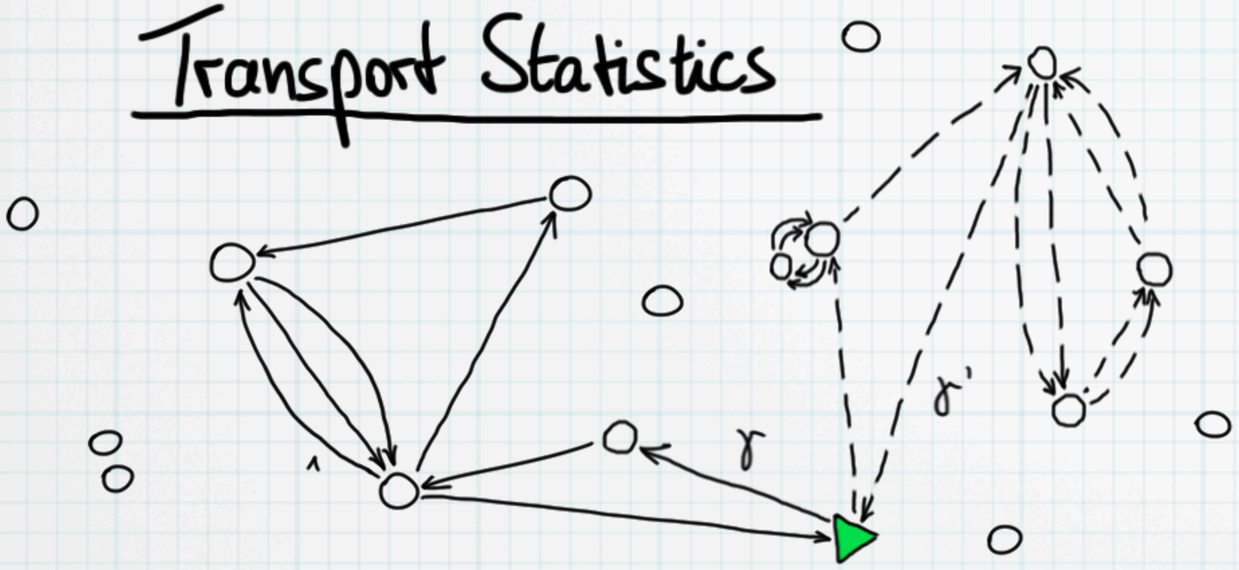
$$= \overline{\sum_\nu |\langle \triangleright | \nu \rangle|^4}$$

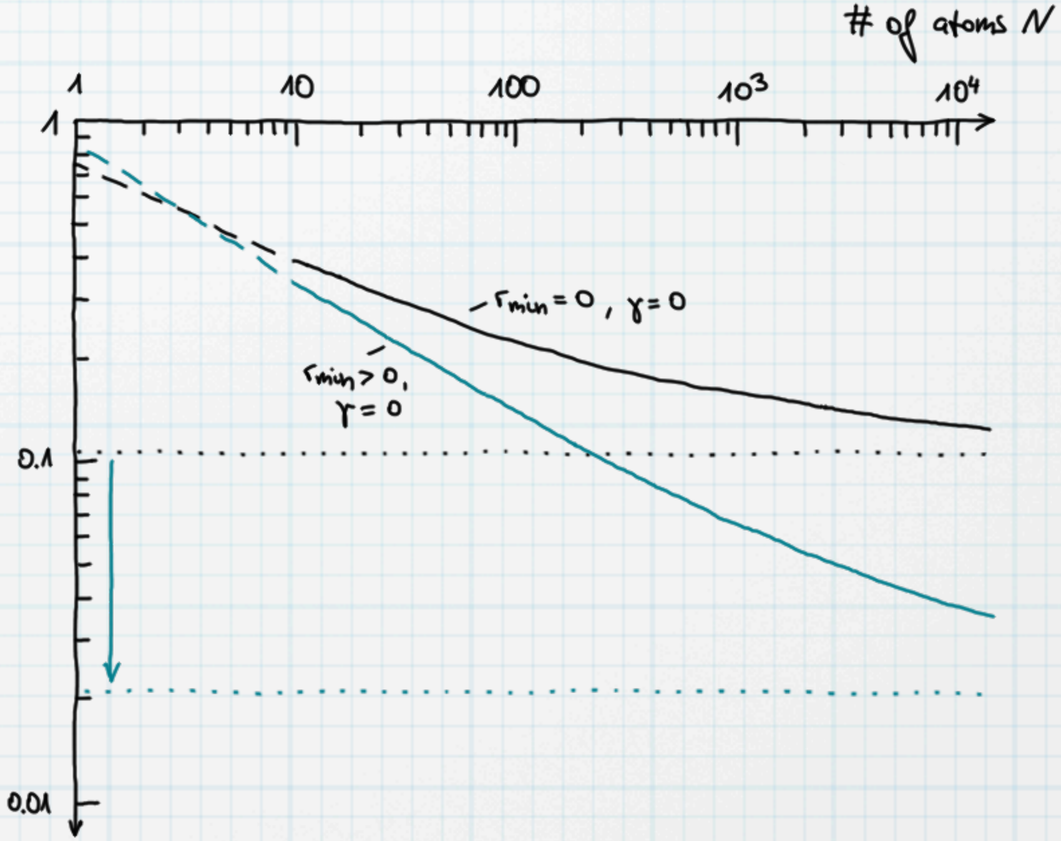
eigenstates of H

- result

$$\overline{P_0} \simeq a + b N^{-c}$$

# Transport Statistics

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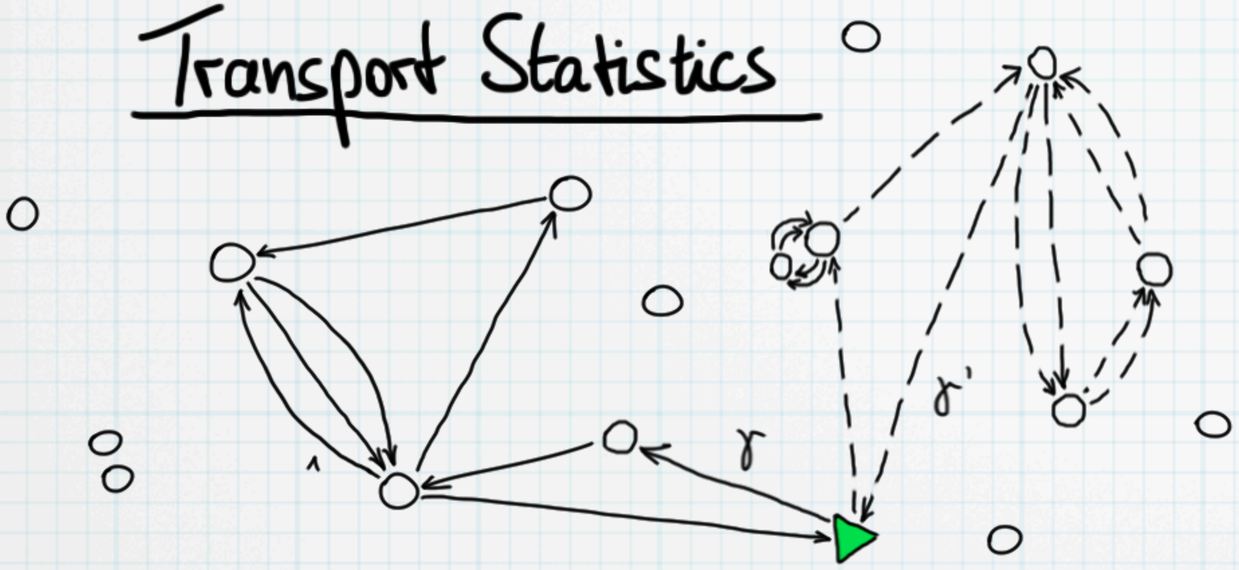


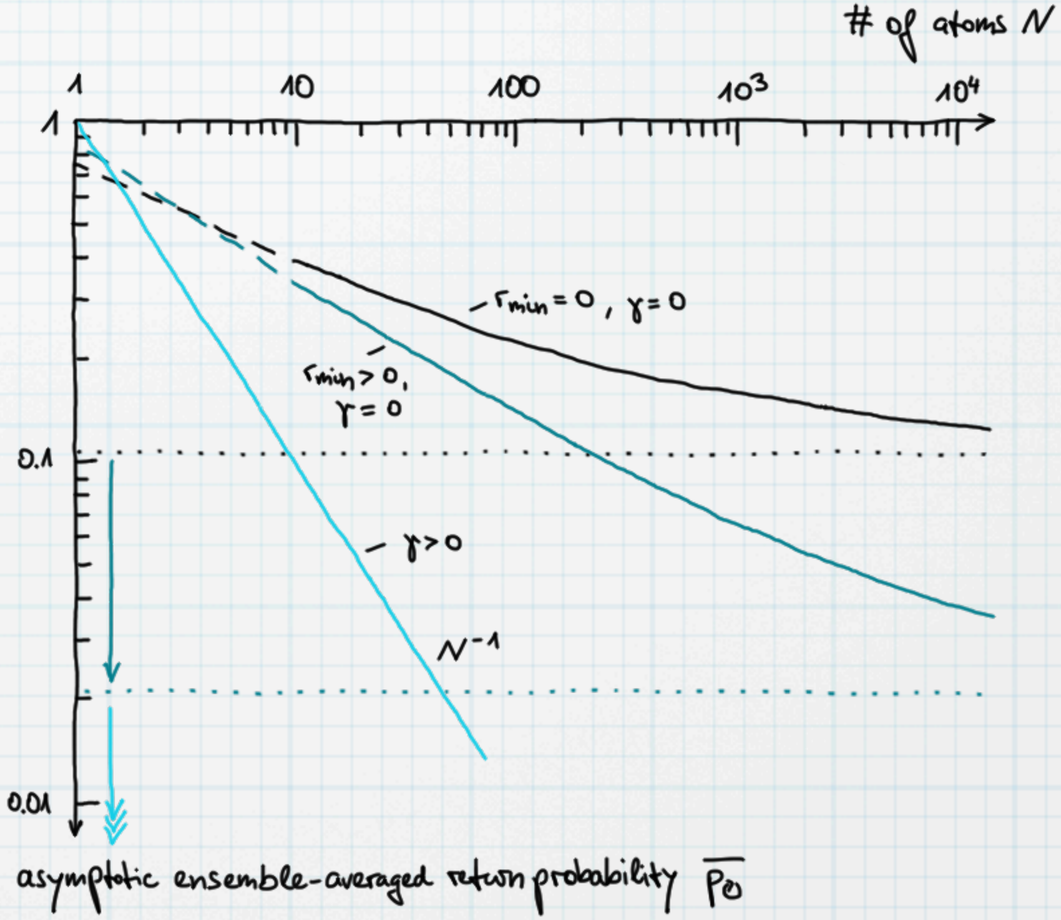


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eigenstates of H

- result

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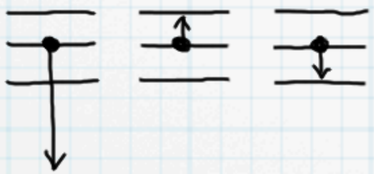
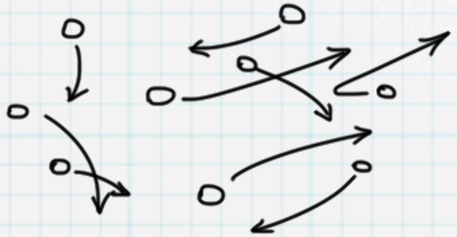
# Decoherence REVISITED

## Rydberg Gas

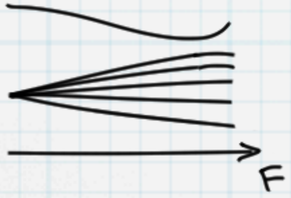
- state redistribution



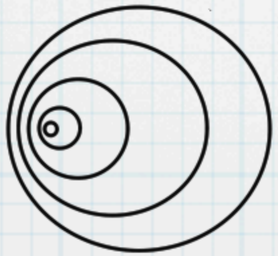
- collisions



- spurious fields

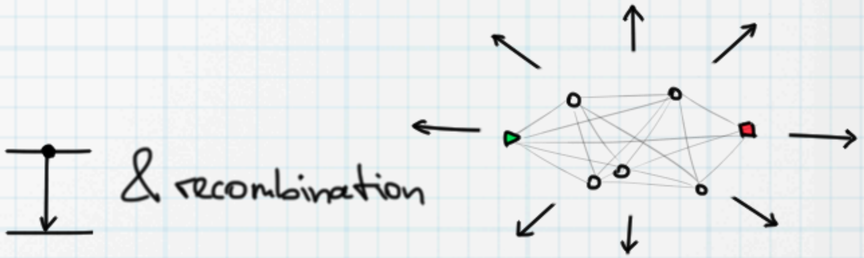



- Doppler effect



## FMO-Complex

- relaxation & recombination

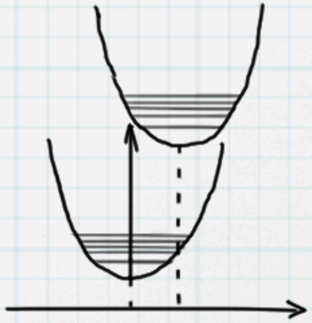


- pigment-protein coupling

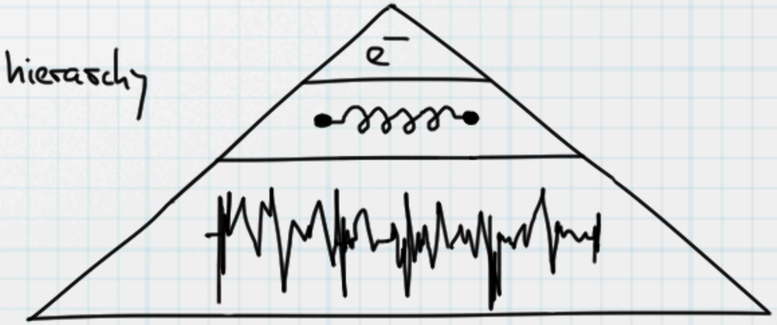
  memory effects,

  correlated fluctuations

  of $E_i$'s & $V_{ij}$'s



- functional hierarchy



# Conclusion

That's all Folks!